

California Influenza Surveillance Dashboard (2009–2020)



Project Overview

This project analyzes influenza activity across California using multiple surveillance systems to better understand **flu trends, severity, and regional impact over time**.

The dashboard combines **Influenza-Like Illness (ILI)** data with **laboratory-confirmed influenza cases** to provide a more complete picture of flu activity from **2009 to 2020**.

Significance

By combining these datasets, this project helps answer:

- Can **ILI trends predict flu outbreaks**?
- How do **confirmed cases compare to symptom-based trends**?
- Which regions experience the **highest flu burden**?

Data Sources & Definitions

1. Influenza-Like Illness (ILI)

ILI represents patients showing **flu-like symptoms**, such as:

- Fever
- Cough
- Sore throat

Key metric:

- **Percent ILI** = % of total patients with flu-like symptoms

Important:

ILI is an **early indicator**, not a confirmed diagnosis.

2. Clinical Laboratory Data

Clinical labs report **confirmed influenza cases** from healthcare settings.

Includes:

- Number of positive cases
- Number of specimens tested
- Positivity rate

Represents:

- **Real diagnosed flu cases**
- More accurate than ILI, but still limited to tested individuals

3. Public Health Laboratory Data

Public health labs provide **official surveillance testing and validation**.

Includes:

- Influenza subtypes
- Total influenza counts
- Number of specimens tested

Represents:

- **High-quality, standardized reporting**
- Used for tracking trends at the population level

Respiratory Viruses Included

This project includes multiple respiratory viruses that produce **flu-like symptoms**:

- **Influenza A**
 - Subtypes: H1N1, H3N2, and others
- **Influenza B**
- **RSV (Respiratory Syncytial Virus)**

Note:

These viruses can present similarly, which is why **ILI alone cannot distinguish between them**.

Dashboard Features

- **KPI Metrics**
 - Total influenza cases
 - Average and peak ILI %

- **Trend Analysis**
 - Weekly ILI % vs confirmed influenza cases
- **Regional Comparison**
 - Flu activity across California regions
- **Severity Classification**
 - ILI categorized as Low, Moderate, or High
- **Virus Breakdown**
 - Influenza A vs B trends over time
- **Testing vs Positivity**
 - Relationship between testing volume and infection rates

Key Insights

- **ILI as an Early Signal**

Increases in ILI often align with or slightly precede confirmed influenza spikes.
- **Seasonality**

Flu activity changes from year to year for projecting hospital burden and positive cases.
- **Regional Variation**

Certain regions experience higher and more frequent outbreaks.
- **Strain Dominance**

Influenza A is typically the dominant strain.